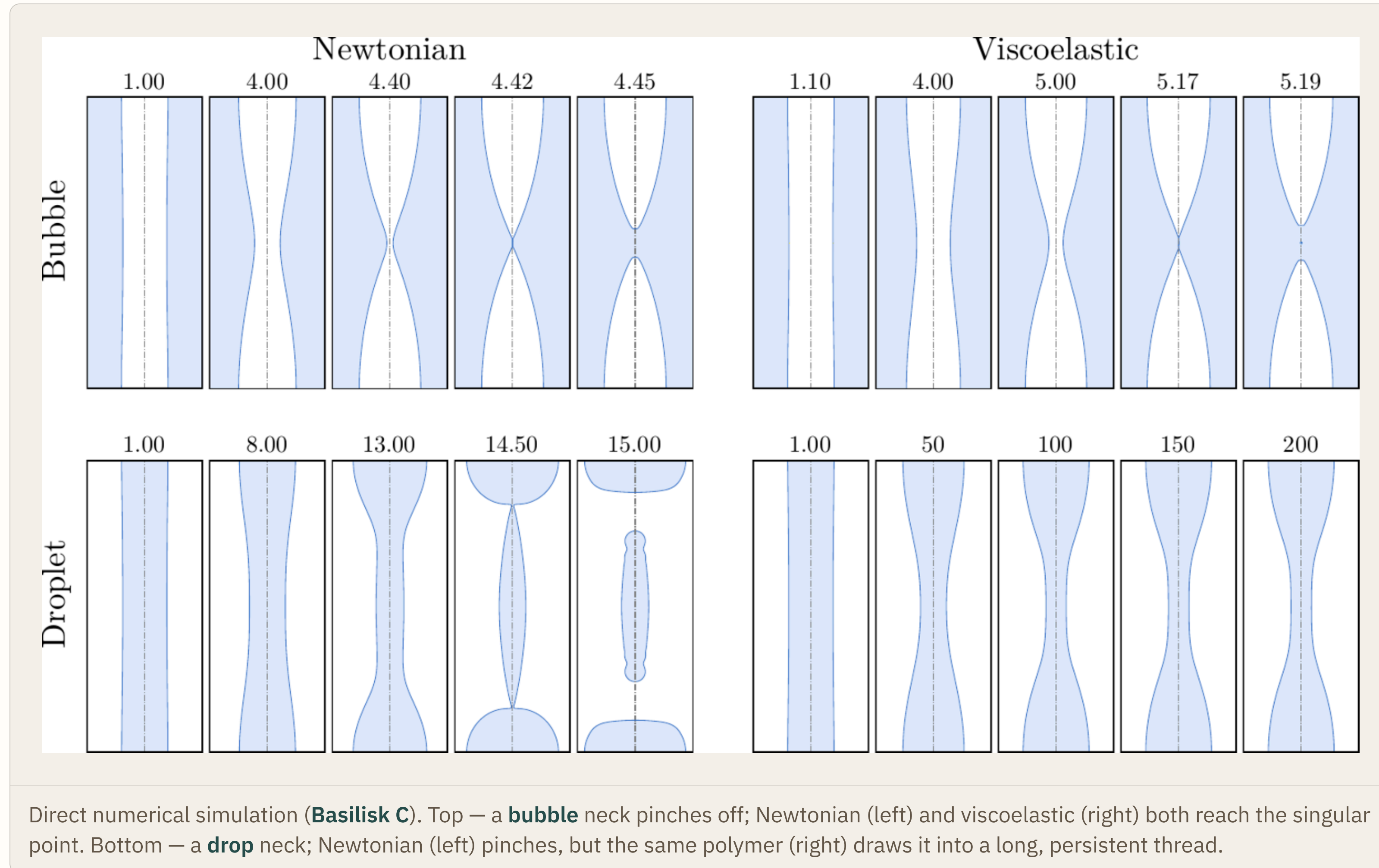
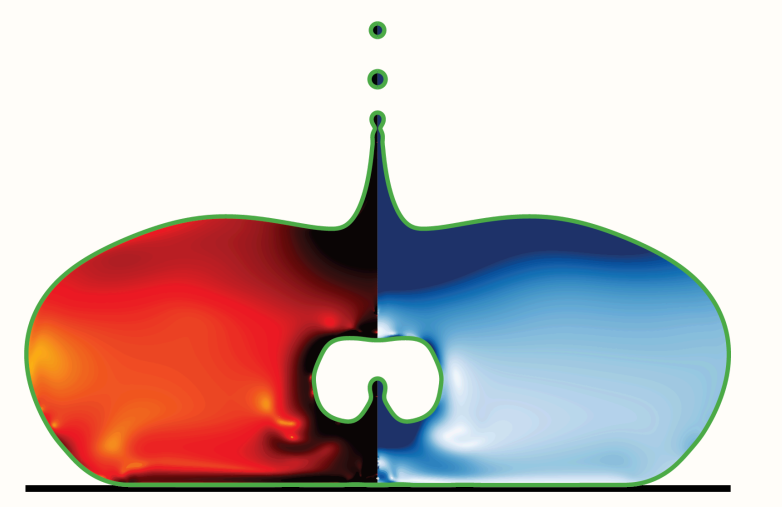


Hydrodynamic singularities

When a free surface pinches off, the flow forgets its past — and a drop and a bubble meet the singularity on opposite terms.

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A **singularity** is where a smooth flow blows up — a finite quantity racing to infinity in a finite time. As a free surface pinches, the neck radius collapses to zero and the dynamics turn **universal and self-similar**: the flow forgets how it began. Add a pinch of polymer and the two textbook cases part ways.

01 What is a singularity?

free-surface flow pinch-off self-similarity DNS · Basilisk

Pinch-off, coalescence, the snap of a sheet or jet — soft-matter flows routinely concentrate geometry, stress and time into a single point. There the neck radius, the curvature and the local strain rate diverge together, and the smooth continuum picture is pushed to its limit.

Close to that point the flow loses memory of how it began. Whatever the nozzle, the liquid or the forcing, the final approach collapses onto the **same self-similar shape** — the fingerprint of a singularity.

The self-similar fingerprint

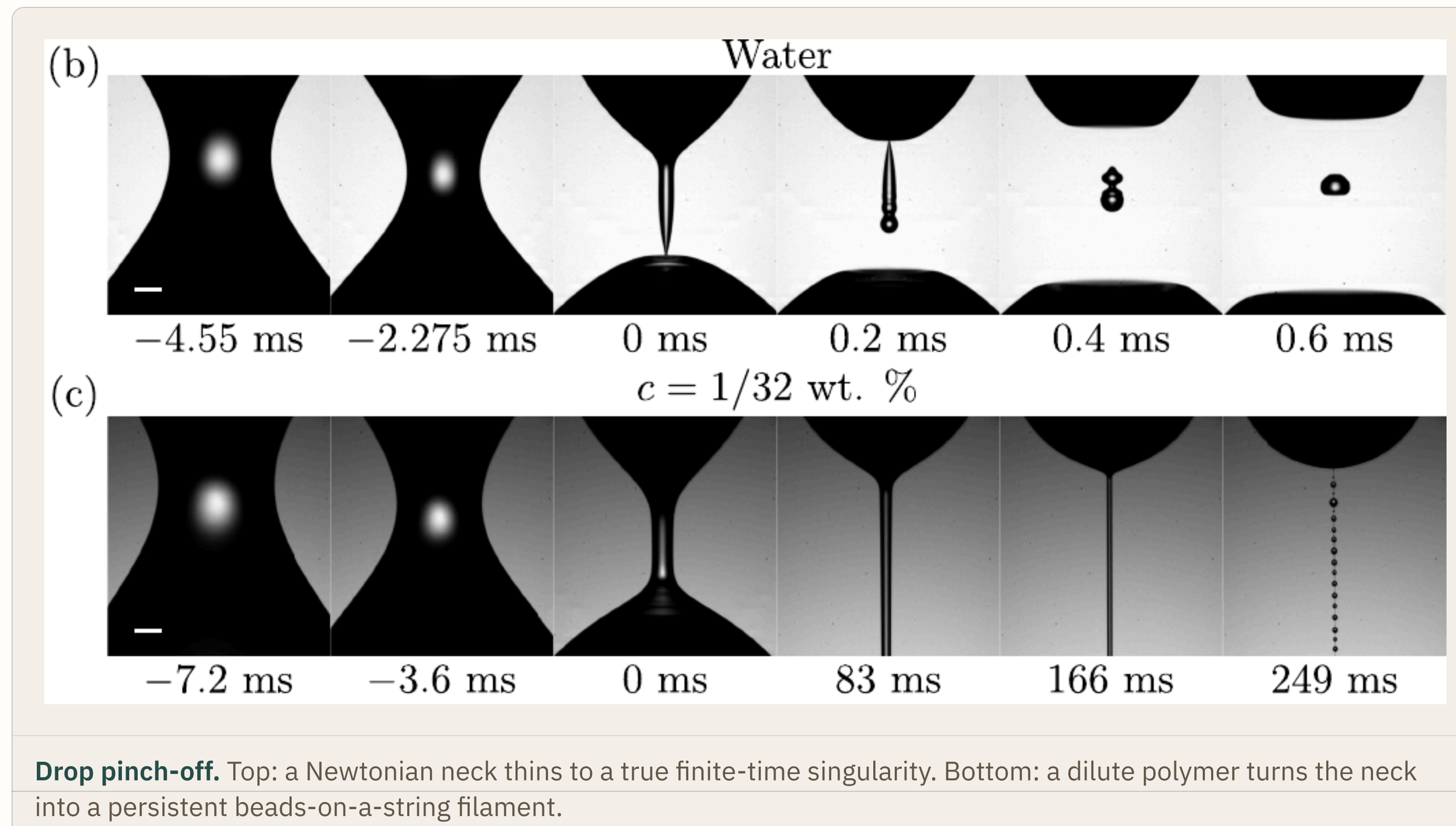
$$h_{\min}(t) \sim (t_0 - t)^\alpha$$

The minimum neck radius vanishes as a power law in the time-to-pinch $t_0 - t$. Rescale every interface by the local neck radius and the shapes collapse onto a single master curve.

02 Drops: elasticity changes the route

A Newtonian drop pinches in finite time: the neck thins to a point and the drop detaches — a clean finite-time singularity.

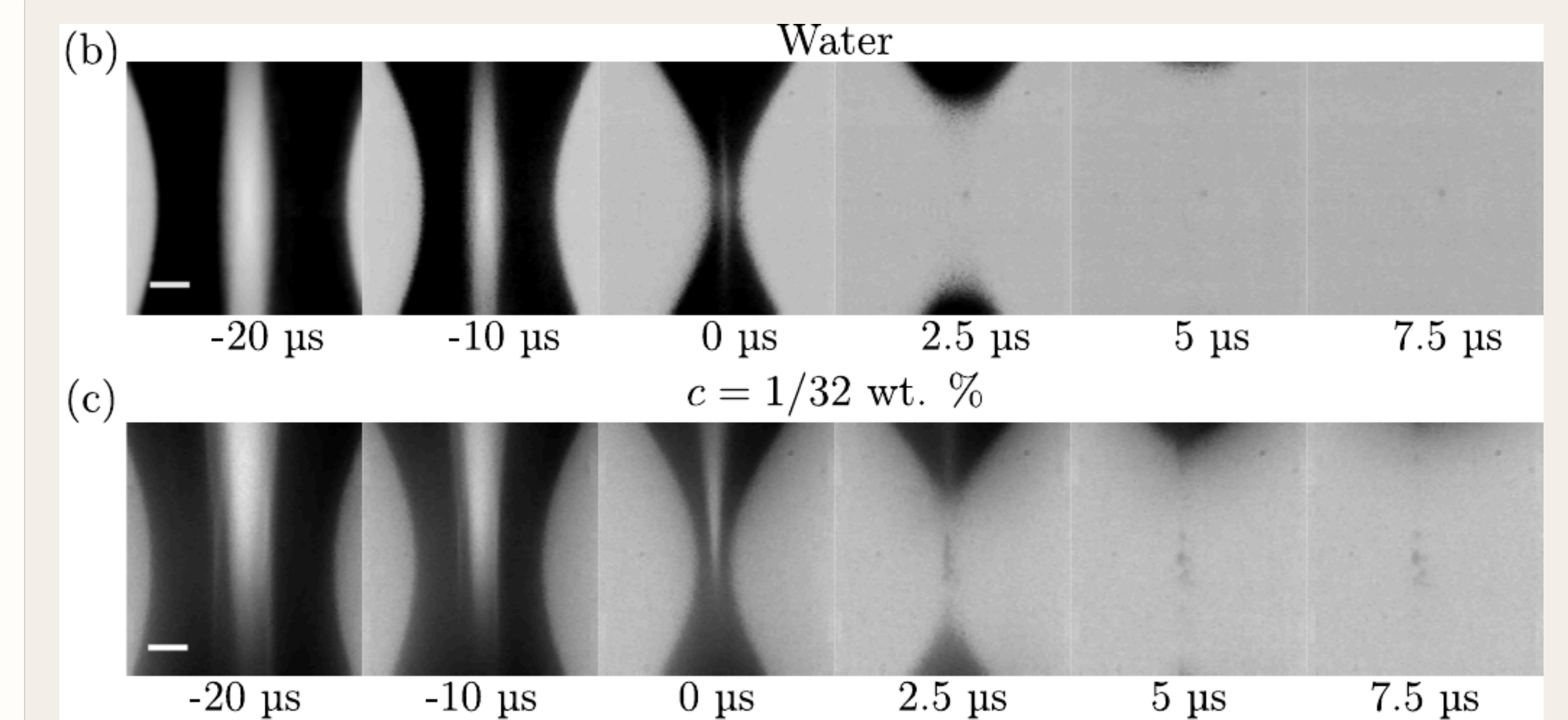
Add polymers and the route changes. As the neck thins, stretching chains build elastic stress along the thread and arrest the collapse. The neck survives as a slender filament that beads up into the classic **beads-on-a-string** — the singularity is postponed, then averted.



03 Bubbles: elasticity leaves the singularity intact

A collapsing bubble neck pinches just as sharply, but the stretching now happens in the liquid *outside* the neck — where the polymer has little chance to organise against the flow.

Add the same polymers and the near-singular sequence looks almost unchanged: the bubble keeps its Newtonian-like approach to the singular point. The **singularity survives**.



Bubble pinch-off. Top: Newtonian. Bottom: viscoelastic. The two near-singular sequences are almost indistinguishable.

→ Polymers spare the bubble singularity

What it tells us

- Two free surfaces, one polymer additive, **opposite outcomes**: elasticity arrests the **drop** singularity but barely touches the **bubble**'s.
- The deciding factor is *where* the fluid is stretched — inside the thinning thread for a drop, outside the collapsing neck for a bubble.
- Singularities are not only where models strain; they are where flows turn **universal**. Knowing which ones elasticity can tame is knowing how to control breakup, sprays and the size of the drops that fly off.

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05 Acknowledgements

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